

PELVIC GIRDLE DEVELOPMENT

The pelvic girdle develops in close proximity to cloacal structures. The anterior portion of the pelvic girdle is intimately associated with the normal development of the caudal part of the urogenital system and the anterior body wall. Studies on the development of the lower limb in avian species and the mouse show that the perineal muscles, including the sphincters, derive from somitic myoblasts that have previously entered the hindlimb bud. Similar to the developmental process seen in the upper limb, somitic myoblasts originating from the dorsoventral edge of the dermomyotome migrate into the somatopleuric mesenchyme of the early lower limb bud. The myoblasts coalesce into a single entity before dividing into dorsal and ventral muscle masses. The cells of the ventral muscle mass then divide again, with one population continuing to develop within the limb and the other migrating ventrally towards the genital tubercle (Valasek et al 2005). In the avian embryo, myoblasts from somites 26–29 enter the limb only, and those from somites 30–33 form both cloacal and limb muscles, whereas those from somite 34 form cloacal muscles solely. Within pelvic girdle development, myoblasts do not attain their position by somitic hypaxial domain extension; all myoblasts migrate into the region. Interestingly, the population of myoblasts giving rise to perineal muscles are innervated by axons derived from neurones located in Onuf's nucleus, whereas those that remain in the limb bud gain innervation from anterior horn cells (Valasek et al 2005).

The bony pelvis forms from two hemipelves, each of which develops from one cartilaginous focus. Ossification of the pelvis starts with the ilium, which undergoes endochondral ossification at 9.5 weeks. Development of trabecular bone in the ilium is evident from about 22 weeks' gestation and a mature trabecular morphology is present at birth (Cunningham and Black 2009a, Cunningham and Black 2009b). It is suggested that the trabecular pattern, previously assumed to develop consequent to weight bearing and bipedal gait, is initiated through genetic patterning plus intrauterine muscular movements which confer biomechanical loading onto the ilium and pelvis.

The development of the superior pubic rami is related to the normal development of the cloacal membrane. In cases of bladder exstrophy (see Fig. 72.11), the superior pubic rami are shortened and do not meet at the pubic symphysis. There is a 4 cm pubic diastasis at birth, which increases to 8 cm at age 10 years, compared with a mean normal width of the pubic symphysis of 0.6 cm at all ages (Wild et al 2011). The ischiopubic segment of the pelvis is 30% shorter, the iliac wings are externally rotated and the acetabuli are retroverted (Wild et al 2011). The mechanisms suggested for this are premature rupture of an abnormally cranially placed cloacal membrane, obstruction of medial migration of somatopleuric mesenchyme, and local apoptosis of ventral mesenchyme (Stec 2011). It is thought that some combination of the three is likely.

The sacroiliac joint can be recognized from 7 weeks; its development is slightly different from that of other synovial joints in that the development of the ilium is ahead of that of the sacrum. Reconsideration of the role of the sacroiliac joint in the transfer of load from axial to appendicular skeleton has been prompted by study of the relative growth of regional surface areas of the ilium from birth to adulthood (Yusuf et al 2013). Cavitation of the hip joint has been reported at 7–8 weeks.

LOWER LIMB DEVELOPMENT

The lower limb is first recognizable as a laterally projecting thickening in the body wall opposite somites 24–29 at stage 13 (28 days); by stage 14, it is closely associated with the wide umbilical cord. During stages 15–17, the limb projects laterally and outgrowth is fairly symmetrical. The core of mesenchymal cells is derived from both somatopleuric and paraxial mesenchyme.

By stage 17, the lower limb still has a flattened foot plate and, although a hip region can be identified, there is no true knee as yet. In stage 18, the lower limb appears to be flexed and abducted at the hip with the knee bent, giving the appearance that the knee is facing laterally. The femur and tibia form in cartilage, and the sciatic nerve extends distally to the tibia by stage 18. Very little skin of the thigh is visible. The soles of the feet face the umbilical cord, and the foot plate has digit rays. During stages 20–23, the digit rays separate and toes are clearly defined by stage 23. The feet can finally touch at stage 21, when the umbilical cord becomes proportionally smaller and the embryo larger. For more details, see O'Rahilly and Gardner (1975).

VESSELS

Arteries

The blood supply to the lower limb is derived from the lateral branch of the fifth lumbar intersegmental artery, which arises from the dorsal root of the umbilical artery passing to the placenta, and continues into the limb bud as the axial artery. The arterial supply to the developing lower limb is thus far less oxygenated than the arterial supply to the upper limb, which receives blood directly from the heart via the subclavian artery. It is possible that the development of several arteries, some of which subsequently regress, in the distal part of the leg in the embryonic period, is a reflection of the lower oxygenation of these tissues. During development, the lower limb undergoes medial rotation and extension, so that the original dorsal (extensor) portion of the limb becomes directed ventrally. This relative movement suggests that very different patterning mechanisms operate in the lower limb compared to those in the upper limb: an early, complex vasculature may be a product of that difference.

There have been relatively few studies on the early development of the vasculature of the lower limb (Senior 1991a,b, Mandell et al 1985, Levinsohn et al 1991). The terminology used to describe the developing vessels is specific and will be used here: descriptions of developmental vascular anomalies retain the original terminology. Table 79.1 describes and illustrates the changes to the axial artery, the new vessels that arise and the vessels that regress, during stages 13–19 (corresponding to weeks 5–7 of development). The mature vascular pattern is established by the end of the embryonic period.

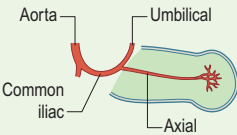
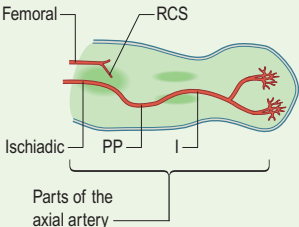
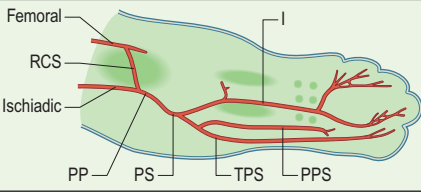
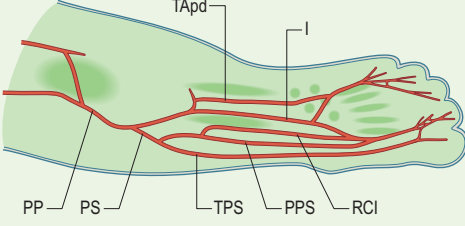
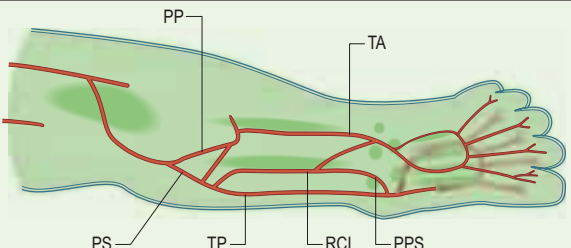
Veins

The superficial veins of the lower limb are derived from the marginal vein which runs along the periphery of the extending limb bud beneath the pre- and post-axial limb borders. The preaxial vein becomes the long saphenous vein, which drains into the femoral vein at the saphenous opening. The postaxial vein becomes the short saphenous vein, which passes deep and joins the popliteal vein. The earliest valves in the short saphenous vein are noted at 13 weeks' gestation. The number of valves increases from 1 to 8 between 13 and 18 weeks' gestation; there are more valves present in the upper than in the lower portion of the short saphenous vein (Czarniawska-Grześcińska and Bruska 2002).

ULTRASOUND ANTENATAL IMAGING

Femur length is routinely measured to assess growth velocity (along with head circumference and abdominal circumference) and to predict the expected date of delivery. The longest axis of the ossified diaphysis is identified with soft tissue noted beyond both ends. The distal epiphysis is excluded (Fig. 79.1). Fetal femur length is related to ethnicity, and appropriate fetal biometry comparison charts are necessary for accurate evaluation; whether such data can be extrapolated to mixed ethnicity fetal populations has been questioned (Ogasawara 2009).

Table 79.1 The development of the arteries of the lower limb during stages 13–19 (corresponding to weeks 5–7 of development), showing the changes to the axial artery, the new vessels that arise and the vessels that regress

Stage of development	Axial artery	Developing arteries	Regressing arteries	Arterial status
11–12	First appearance of the lower limb bud. The lower limb axial artery arises from the fifth lumbar intersegmental branch of the aorta.			
13–16	The axial artery courses through the lower limb bud to the developing digital plate where it branches into a terminal plexus (future dorsal and plantar networks).			
16–17	The axial artery is now described as ischiadic proximally and popliteal profunda (PP) and interossea (I) as it progresses distally.	The external iliac artery, which arises just proximal to the origin of the ischiadic artery, extends distally towards the knee as the femoral artery. A branch, the ramus communicans superior (RCS), passes dorsally to join the ischiadic artery. Perforating branches of the profunda femoris artery initially anastomose with the proximal part of the ischiadic artery.		
18	Elongation of the axial artery forms the popliteal superficialis (PS) between the popliteal profunda and interossea arteries.	Two new vessels arise from the popliteal superficialis artery. These are the tibialis posterior superficialis (TPS) and the peronea posterior superficialis (PPS). Both vessels join the plantar network. There are now three arteries in the leg.	The proximal portion of the ischiadic artery starts to regress. Parts persist in the adult as the inferior gluteal artery and the arteria comitans nervi ischiadici.	
19		Two further vessels arise. They are the tibialis anterior pars distalis (TApd) from the interossea artery and the ramus communicans inferior (RCI) from the peronea posterior superficialis. The original plantar network is remodelled into a medial plantar artery (from the distal part of tibialis posterior superficialis) and a lateral plantar artery (from the distal continuation of the confluence of the interossea, ramus communicans inferior and the peronea posterior superficialis) which together form the plantar arch. There are now five arteries in the leg.		
20		The arteries in the leg at this stage are the tibialis anterior (TA), tibialis posterior and the ramus communicans inferior. The latter is confluent with the peronea posterior superficialis in the tarsal region and becomes the fibular (peroneal) artery. The plantar arch is now supplied solely by the tibialis posterior. There are now three arteries in the leg.	Popliteal profunda, interossea artery and peronea posterior superficialis arteries.	

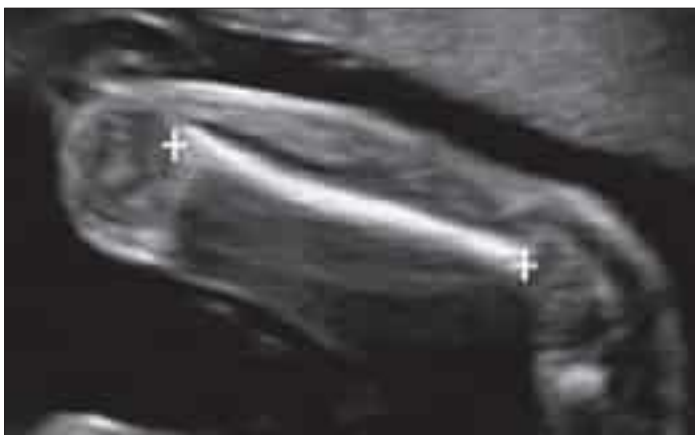


Fig. 79.1 An ultrasound image showing femur length. (With permission from To M, Pereira S 2015 Routine fetal anomaly scan, in Coady AM (ed) Twining's Textbook of Fetal Abnormalities, 3rd edition, Churchill Livingstone.)

Identification of short femur length is associated with a higher risk of chromosomal abnormalities (Mathiesen et al 2014) and a significantly higher likelihood of the fetus being small for gestational age and at risk of early preterm delivery (Mathiesen et al 2014, Özlü and Ozcan 2013). It has been estimated that identification of isolated short femur length below the tenth or fifth percentile at mid-gestation is associated with more than a three-fold risk of the development of fetal growth retardation and increased risk of early preterm birth before 37 and 34 weeks (Goetzinger et al 2012). Although the cause of this relationship must be multifactorial, it is suggested that changes in placental and chronic fetal hypoxia may lead to a fetal adaptive response in which highly oxygenated blood is diverted to the brain and heart at the expense of the extremities.

NEONATAL LOWER LIMB

In the neonate, the pelvis is cone-shaped. The transverse diameter of the true pelvis is 2.2 cm, its anteroposterior diameter is 2.8 cm, and the length between the inlet and outlet is 2 cm. The sacrum is proportionately larger than in the adult and the sacral promontory is higher. When walking begins, the sacrum descends between the ilia and the promontory develops. The ilia, ischia and pubic bones are variably ossified at

birth; they meet at the acetabulum, which in the neonate is cartilaginous, relatively large and shallow.

The lower limbs are underdeveloped in the neonate when compared to the upper limbs. They are retained in a flexed position and the lower leg is proportionately shorter than the thigh. Although the legs appear to be bowed, the tibia and fibula are straight; the illusion of bow legs is caused by the shape of the soft tissues and the slightly more advanced development of the lateral head of gastrocnemius compared to its medial head. The femoral neck is much shorter and forms an acute angle with the shaft. The latter is quite straight because the adult curvature is acquired with walking. The head of the femur is larger than the acetabular fossa and nearly one-third remains external, which means that the ligament of head of femur is relatively very long. Dislocation of the hip joint is relatively easy; the femoral head can be removed from the acetabular fossa laterally but not posteriorly. The calcaneus and the talus have an ossification centre at birth, and a centre is present in the cuboid in half of neonates.

The muscles of the lower limb are much less developed than those in the upper limb. The fetal position often assumed by postnatal babies keeps the thighs in continuous abduction, stretching the adductors. The muscles that will be used for walking are weak; the lack of gluteal development, in particular, gives the typically diminutive buttocks of the neonate.

In neonates, the feet are usually inverted and they have a greater degree of dorsiflexion, caused by the relatively greater area of the trochlea of the talus. Plantar flexion is limited, in part reflecting the shortness of the extensor muscles of the foot. At birth, the footprint outlines the whole plantar surface, reflecting the deposition of subcutaneous fat beneath the longitudinal and transverse arches, and so most babies appear flat-footed.

DEVELOPMENTAL ANOMALIES OF THE LOWER LIMB

The categories of limb defects described by Swanson (1976) are given on page 796. Although devised for the upper limb, they similarly apply to the lower limb.

Developmental dysplasia of the hip (previously known as congenital dislocation of the hip) refers to an abnormal configuration of, or relationship between, the femoral head and the acetabulum that occurs either before or after birth in approximately 1 in 100 live births with a female: male ratio of 6:1. (In 2% of cases, hip dislocation is not evident at birth but manifests in the first few months of the life.) It covers a continuum of disorders that ranges from shallowness of the acetabulum to instability and subluxation of the femoral head and, ultimately, to frank dislocation. Acetabular anteversion may occur (Li et al 2009). The aetiology is considered to be multifactorial and is associated with first pregnancies, suggesting that both maternal and uterine musculature restricts fetal movement and puts postural strain on the fetal hips.

Developmental dysplasia of the hip is seen more frequently in breech delivery, especially if the child's knees are extended. The incidence is similar in preterm and term infants born in the breech position (Quan et al 2013). The left hip is more frequently affected than the right, possibly because, in breech presentation, the fetus lies with the right shoulder anterior and the left thigh closest to the maternal sacrum. The physiological effects that cause the maternal ligaments to become temporarily lax prior to delivery are also considered to affect the fetus and to contribute to laxity of the hip capsule. Risk factors are breech delivery, female sex, a positive family history and clicking hips at clinical examination (de Hundy et al 2012). Ultrasound is beneficial in the early detection and assessment of children with developmental dyspla-

sia of the hip; magnetic resonance imaging (MRI) may be beneficial in identifying the underlying bony and soft tissue anatomy.

The collodiaphysal angle (CDA; neck-shaft angle, Mikulicz angle) is the angle between the long axis of the femoral shaft and the head and neck of the femur; the normal range varies between 125° and 140° (see Fig. 81.10). Where the CDA is enhanced, usually above 135°, a coxa valga deformity occurs. Where CDA is decreased (usually defined as less than 120°), a coxa vara deformity occurs; the mechanism is thought to be a failure of medial growth of the physal plate. The resulting lower limb is shortened and abduction is restricted. In both coxa valga and coxa vara, the long-term unequal weight distribution may result in excessive wear of articular cartilage.

The femur normally aligns on the tibia obliquely, creating an angle of 174° facing laterally. A medial angulation of the leg in relation to the thigh is a deformity called genu varum, whereas a lateral angulation is called genu valgum. The hip-knee (femoral-tibial) angle in children is +3.6° (varum) between the ages of 1 and 2 years, and -2.5° (valgum) between the ages of 2 and 3 years. The prevalence of lower limb deformities physiologically decreases after 5 years of age; the angle is +0.3° after the age of 7 years, which is within 1° of adult reference values (Sabharwal and Zhao 2009). Both deformities cause unequal weight distribution that hastens destruction of knee cartilages; persistence of genu varum/valgum until late childhood may require correction in order to prevent arthrosis. (For further reading, see Goldman and Green (2010).)

The patella may be absent or hypoplastic, a malformation that may be isolated or may be associated with other defects (Mizobuchi et al 2007).

Congenital talipes equinovarus, or club foot, derives its name from a combination of *talus* and *pes*, together with terms describing an elevated heel resembling that of a horse (*equino*), which is also turned inwards (*varus*). It is a common neonatal anomaly and occurs in approximately 1 per 1000 live births, with males affected twice as often as females. There is some degree of inheritance, so it is not entirely an effect of intrauterine positioning or of oligohydramnios (abnormally little amniotic fluid). Both bones and soft tissues are affected and it is difficult to tell which are primary effects and which are secondary. A number of theories have been proposed to explain the underlying pathogenetic mechanism(s), including abnormal tendon and ligament attachments, defective development of the talus and delayed muscle maturation. One theory is that the talus undergoes defective development and that all the other deformities arise as a consequence of this initial defect. The talus is decreased in size by up to 25%. It has a foreshortened neck and decreased body/neck angle, and the subtalar facets are medially rotated. The navicular is small and medially deviated relative to the talus. The calcaneus is also small and shows varus displacement and equinus tilt; the anterior facets correspond to those of the talus (Barlow and Clarke 1994). The entire affected foot and calf are smaller than their normal counterparts. The foot is inverted and supinated, and the forefoot is adducted. The heel is small, rotated inwards and elevated. The calcaneus is inverted beneath the talus. MRI analysis of the lower legs of patients who had been treated for idiopathic congenital talipes equinovarus showed that leg muscle volume was consistently reduced (Duce et al 2013). Treatment varies between splintage and repeated complex surgery, which reflects the highly variable severity of the condition and individual response to therapy.

'Flat feet' are common in childhood. The majority are 'flexible' and simply related to posture, whereas 'rigid' flat feet are caused by structural abnormalities. The description of 'rocker bottom' foot is associated with many congenital syndromes. The foot is flat and rigid, and the plantar surface appears curved with the apex of the curve at the mid-tarsal joint. The condition shows an equinus position of the hindfoot. The talus may be vertical and palpable on the plantar surface. A majority of infants with this condition have neurological abnormalities.

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